

Bridging AI Ethics and Requirements Engineering in Healthcare: Triangulating Evidence from Literature, Experts, and Online Communities

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Abstract

With the growing number of ethical frameworks for AI in healthcare, a persistent gap remains between high-level principles and implementable software requirements. Existing guidance often treats “stakeholders” as a single group, obscuring distinct priorities across roles and settings. We address this gap through methodological triangulation that synthesises ethical requirements from three complementary sources: (i) a systematic mapping study of 39 primary studies, (ii) semi-structured interviews with 17 practitioners, and (iii) large-scale mining of 22,296 Reddit threads using a multi-agent LLM pipeline. Triangulation surfaces (a) convergent priorities that recur across sources (e.g., privacy and safety), (b) complementary insights that differentiate stakeholder concerns, and (c) unique deployment issues that individual methods miss. We show how cross-source agreement and disagreement can be used to derive confidence levels that support requirements prioritisation and justify trade-offs. Building on these findings, we outline a preliminary ethics assistant design that translates triangulated evidence into practical requirements prompts and checks for engineers. Overall, this paper advances a generalisable approach for bridging abstract AI ethics and concrete requirements engineering in healthcare, and sets an agenda for evaluating triangulation-driven tool support in real development workflows.

Keywords

Responsible AI, AI Ethics, Software Engineering, Healthcare AI, Requirements Engineering

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1 Introduction

AI is rapidly reshaping healthcare software, but a recurring failure point is not model performance; it is the translation of ethical principles into implementable software requirements [1]. This translation gap is evident in practice: among over 1,000 AI-enabled medical

devices registered with the U.S. FDA, only 3.6% report patient demographics, and only 28% undergo multi-institutional validation [2]. In parallel, documented algorithmic bias has affected over 200 million U.S. citizens, with mortality rates varying by 24% across racial groups in a widely deployed risk prediction system [3]. These figures indicate that current ethical frameworks and guidelines have not yet become routine, checkable requirements in everyday software engineering (SE) workflows. A key reason is that the evidence used to derive “ethical requirements” is fragmented across methods that each provide only a partial view. We see three dominant approaches in prior work:

Literature syntheses (SLRs and mapping studies). Reviews consolidate academic and policy perspectives on ethical concerns (e.g., privacy, safety) and compile mitigation strategies [4]. However, they are bound by what is published: they over-represent idealised solutions and under-represent deployment constraints, organisational incentives, and the practical work of integrating ethics into requirements and verification.

Practitioner interviews. Interview studies provide contextualised accounts of ethical concerns in day-to-day practice and expose organisational and workflow constraints that are often absent from published guidelines [5–7]. Yet interviews are limited by sample size and geographic scope and are vulnerable to social desirability bias, as participants may align their responses with perceived professional expectations rather than the concerns they discuss informally with peers [8].

Mining online discussions. Large-scale analysis of online forums can capture candid, experience-driven concerns at breadth, including deployment issues that may not arise in formal studies [9–11]. However, online data presents representativeness questions and is rarely linked back to other sources in ways that support systematic requirements engineering (RE).

Despite substantial work in each stream, the field still lacks a clear answer to a requirements-centric question: *which ethical concerns and mitigations are robust across evidence sources, and which are artefacts of a particular method or setting?* Without this, engineers have limited guidance on what to prioritise, how to justify trade-offs, and what constitutes sufficient evidence for turning an ethical principle into an implementable requirement.

Our approach: triangulation for ethical requirements. We address this gap using *methodological triangulation* [12], treating literature, interviews, and online discussions as complementary lenses on

the same phenomenon. Triangulation is particularly appropriate for complex socio-technical systems in requirements engineering, where no single method captures all relevant perspectives [13]. However, to our knowledge, it has not been applied to derive and validate ethical requirements for healthcare AI by systematically integrating (i) published frameworks and studies, (ii) practitioner accounts, and (iii) large-scale peer discussion.

Concretely, we synthesise ethical requirements from three empirical sources and then cross-validate them:

Systematic mapping study (39 primary studies): We identify ethical dimensions, mitigation strategies, and implementation challenges reported in academic and guideline literature [1]. We find that 67% of proposed mitigation strategies lack empirical evaluation, reinforcing a practical gap in the proposed ethics solutions.

Semi-structured interviews (17 practitioners): We interview healthcare professionals, healthcare researchers, and software engineers across China, Australia, and India. These interviews surface role-specific priorities and inform a stakeholder-specific “If-Then” rule set intended to support requirements formulation. The depth of interviews is valuable, but their scale and potential desirability effects motivate an additional, broader evidence source.

Large-scale mining (22,296 Reddit threads): We apply multi-agent LLM-based annotation to extract concerns expressed in candid peer discussions on Reddit. This provides breadth and reveals deployment issues that practitioners discuss among peers but may not raise on other forums, such as interviews.

By integrating these sources into a unified triangulation pipeline, we distinguish (i) *convergent* concerns that recur across methods, (ii) *complementary* concerns that vary by stakeholder role, and (iii) *unique* concerns that appear primarily in deployment-facing discourse. We further show how cross-source agreement and disagreement can be used to assign *confidence levels* to candidate requirements, supporting requirements prioritisation and making underlying evidence explicit. Building on these findings, we outline a preliminary ethics assistant design that operationalises triangulated insights into practical requirements support.

Contributions. This paper makes the following contributions: (i) a triangulation method for deriving ethical requirements for healthcare AI from literature, interviews, and online discussions; (ii) an integrated synthesis of ethical concerns and mitigations that separates convergent, complementary, and unique findings across evidence sources; (iii) an evidence-grounded approach to confidence assessment to support requirements prioritisation; and (iv) a preliminary design for tool support that translates triangulated evidence into engineer-facing prompts and checks.

Paper organisation. Section 2 presents background and related work. Section 3 describes three empirical studies and our triangulation methodology. Section 4 reports findings across ethical dimensions. Section 5 discusses implications and next steps. Section 6 outlines threats to validity and Section 6 concludes.

2 Background and Related work

2.1 Operationalizing Ethical AI in Healthcare

Efforts to govern AI in healthcare vary across international organizations, regulatory bodies, and professional associations. A broad set of guidance documents states the ethical principles for AI in

healthcare. The WHO emphasises human-rights-centred governance [14], the EU AI Act defines risk-based regulatory requirements [15], and the NIST’s AI Risk Management Framework addresses trustworthiness across the AI lifecycle [16].

Despite the abundance of guidance, translating principles into practice remains difficult. Bürger et al. found that 75% of major ethical guidelines provide only high-level principles with minimal practical guidance, leaving engineering teams without actionable methods to implement concepts like fairness or accountability [17]. Goirand et al.’s scoping review of ethics implementation in healthcare AI similarly concluded that while principles are frequently discussed, tools and processes for testing and upholding ethical requirements are largely absent [18]. Other systematic reviews reinforce this gap: Li et al. identified twelve ethical issues across 45 healthcare AI documents yet noted the need for practitioner engagement [19], while Atlam et al. mapped 127 AI ethics studies and found a lack of empirical evidence supporting ethical principles in practice [20]. These findings highlight a persistent gap between what frameworks describe and what practitioners can implement.

2.2 Stakeholder Perspectives on Healthcare AI

It is essential to understand how different stakeholders perceive ethical concerns, as this helps with deriving actionable requirements. Prior interview studies reveal that the perspectives vary significantly by role. Elgin and Elgin found that clinicians express concerns around transparency, equity, and accountability when using AI-driven decision support, revealing tensions between efficiency gains and fairness considerations [21]. Sonawane et al. identified recurrent concerns about data privacy and the potential for AI to exacerbate care inequalities among healthcare professionals [22]. Ramadan et al. reported that nurses fear that AI will undermine the nurse-patient relationship [23].

However, most studies focus on a single stakeholder group in healthcare, typically clinicians, while software engineers who build these systems remain underrepresented [24]. This is a significant gap, as software engineers make design decisions that embed ethical trade-offs into system behaviour.

2.3 Mining Online Discussions

Online repositories have emerged as valuable sources for requirements elicitation. Ferrari and Spoletini’s study with 58 requirements analysts found that mining app stores uncovered novel features, with up to 42% of requirements inspired by online sources covering functionality beyond traditional interview-based elicitation [25]. Iqbal et al. demonstrated that Reddit can serve as a source for software requirements, capturing user needs expressed in naturalistic settings [26]. Tjikhoeri et al. analysed ethical concerns in app reviews, showing that users discuss privacy, fairness, and transparency issues through informal channels [27].

These studies have shown that online discussions capture concerns as they arise in practice and complement resource-intensive methods such as interviews. However, mining faces challenges including representativeness, noise and annotation reliability [28]. Recent advances in LLMs offer the potential for automated analysis at scale, but the application of LLMs to the extraction of ethical requirements from healthcare discussions remains underexplored [29].

3 Methodology

This study implements methodological triangulation to synthesize stakeholder perspectives on ethical requirements for healthcare AI. We combine three complementary empirical approaches: a systematic mapping study of academic literature and industry guidelines, a semi-structured interview with practitioners, and a large-scale mining of online discussions. Figure 1 illustrates our overall approach.

3.1 Research Questions

Our study aims to address the following research questions: **RQ1**: What ethical concerns are associated with generative AI in healthcare? **RQ2**: How do these concerns compare across academic literature, practitioner interviews, and online discussions? **RQ3**: How can these triangulated findings support the design of tools for ethical requirements elicitation?

3.2 Systematic Mapping Study

We conducted a systematic mapping study following established guidelines [30, 31] to identify ethical concerns and mitigation strategies in academic literature and industrial guidelines as discussed in detail in our paper which is under review [1]. The mapping study also established the ethical dimensions used in our analysis for triangulation.

3.2.1 Search Strategy. We searched six databases (IEEE Xplore, ACM Digital Library, ProQuest, Web of Science, Wiley Online Library, and ScienceDirect) using search strings combining terms related to large language models (e.g., “large language model,” “generative AI,” “GPT”), ethics (e.g., “ethic*,” “bias,” “fairness,” “privacy”), and healthcare (e.g., “healthcare,” “medical,” “clinical”). We adapted the search string to each database’s syntax requirements. We also conducted targeted searches of the websites of major AI ethics bodies to capture industry and governmental guidelines.

3.2.2 Selection Criteria. Inclusion criteria: (1) peer-reviewed studies or grey literature discussing ethical concerns in generative AI; (2) written in English; (3) full text available; (4) published between 2020-2024. Exclusion criteria: (1) studies not addressing ethical concerns; (2) papers under four pages; (3) non-research publications such as posters and opinion papers.

3.2.3 Data Extraction and Results. The initial search yielded 5,013 records. After removing duplicates and applying selection criteria through title, abstract, and full-text screening, followed by forward snowballing, we selected 39 primary studies. Two researchers independently extracted ethical concerns, mitigation strategies, and implementation challenges from each study. The extracted data were coded and organized into themes through discussion.

3.2.4 Ethical Dimensions Identified. From the mapping study, we identified five recurring ethical dimensions across both academic literature and industry guidelines: Privacy (protection of personal health information), Safety (minimizing risks and ensuring reliable operation), Bias (fairness and equity in AI outputs), Transparency (explainability of AI decisions), and Accountability (responsibility for AI-driven outcomes). These dimensions appear consistently

across major frameworks, including the EU AI Act, NIST AI RMF, IEEE Ethically Aligned Design, and Microsoft Responsible AI Standard [16, 32–34]. We adopted these five ethical dimensions in our triangulation approach to enable systematic comparison across data sources.

3.3 Semi-Structured interviews

We conducted semi-structured interviews to capture practitioner perspectives on ethical priorities and challenges in healthcare AI.

3.3.1 Participant Recruitment. We employed a multi-channel recruitment strategy, advertising our study on social media platforms including LinkedIn, WeChat, and X (formerly Twitter), as well as through professional networks. We recruited 17 participants (8 female, 9 male) across three stakeholder groups: healthcare professionals (n=5, including clinicians, nurses, and radiologists), healthcare researchers (n=6, conducting AI-related health research), and software engineers (n=6, developing healthcare AI systems). Participants were recruited from China, Australia, and India and worked in diverse settings, including IT companies, clinics, hospitals, and universities, with 2-24 years of professional experience in their respective fields. Our study was covered by an Ethics Application at our university.

3.3.2 Interview Design. Interviews were conducted between February and July 2025, both via Zoom and in person, lasting 30-45 minutes each. We conducted three pilot interviews (one participant from each group) and made minor refinements based on feedback. The final interview protocol covered three areas: (1) *Practical use scenarios*: how participants use healthcare AI tools in clinical practice, research, or software development, and the benefits they experience; (2) *Current challenges and limitations*: specific incidents and unresolved issues encountered in practice; and (3) *Ethical considerations*: perspectives on the five ethical dimensions. For the ranking exercise, we provided definitions and healthcare-specific examples for each dimension to ensure consistent interpretation, then asked participants to rank these dimensions by importance and explain their reasoning. We acknowledge that providing predefined dimensions may influence responses; however, this approach enabled systematic comparison across participants and alignment with our triangulation framework.

3.3.3 Thematic Analysis. The interviews were transcribed and analyzed following thematic synthesis guidelines [35]. We applied a five-step process: (1) initial reading of transcripts with notes on recurring ideas; (2) segmenting data aligned with research questions; (3) open coding using hashtags (e.g., “AIUseCase,” “TransparencyIssue”); (4) developing sub-themes by merging related codes; and (5) grouping sub-themes into higher-level themes. Two researchers independently coded transcripts, with codes iteratively refined through discussion until consensus was reached.

3.4 Online Discussion Mining

We extracted Reddit discussions to capture stakeholder concerns at scale, complementing the depth of interviews with the breadth of practitioner discussions.

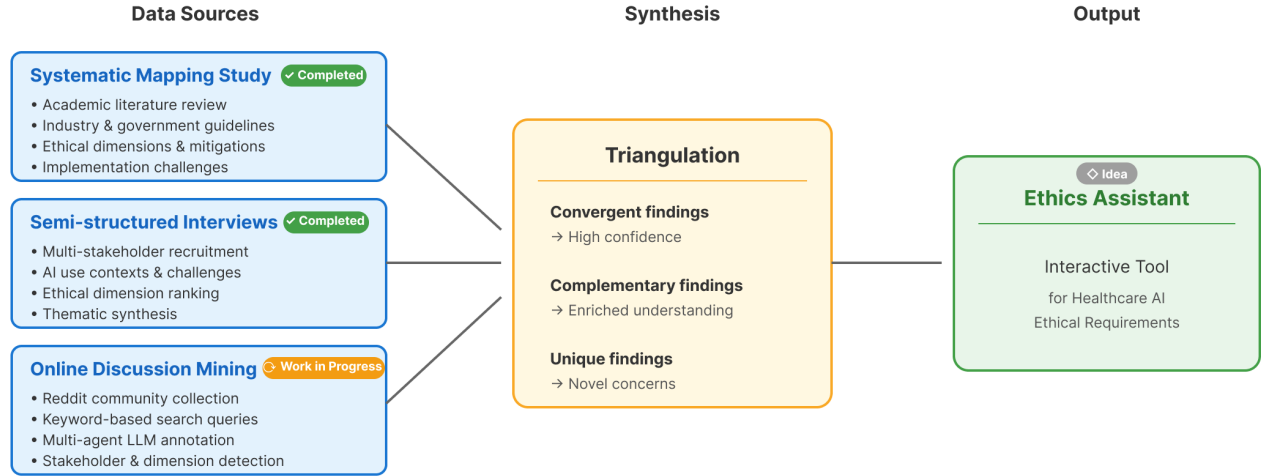


Figure 1: Triangulation approach

3.4.1 Subreddit Selection and Query Design. Reddit is organized into topic-focused communities called *subreddits* (e.g., *r/medicine*), where users create *threads* and respond through *comments*. Unlike X or Facebook where content flows through personal networks, Reddit groups discussions by topic, making it suitable for identifying domain-specific stakeholder conversations [11].

We targeted public subreddits aligned with three stakeholder groups: Healthcare Professionals (HCP), Healthcare Researchers (HCR), and Software Engineers (SEng). A subreddit was included if it: (1) targeted one of these audiences, (2) had healthcare AI content, and (3) had more than 30 posts per month. This resulted in 33 subreddits, including *r/medicine*, *r/healthIT*, *r/MachineLearning*, and *r/bioinformatics*, etc. We collected threads from 01/2022–08/2025.

We constructed 86 search queries by combining AI, healthcare, and ethics-related keywords that represent the five dimensions. This combinatorial approach was necessary because ethical discussions are often implicit. We piloted queries on small samples and iteratively refined them following Shahin et al. [36].

3.4.2 Data Processing. Using Reddit’s OAuth API, we executed search queries across the 33 targeted subreddits. For threads with comments, we collected up to 1,000 comments using Reddit’s comments with the sort=“top” parameter to capture comments with the most positive votes. Our initial queries retrieved approximately 110,000 threads. We then applied filters: removed exact duplicate threads based on ID, kept only English-language threads using language detection, and removed threads posted by deleted or suspended accounts. The final dataset contained 22,296 unique threads.

3.4.3 Multi-Agent LLM Annotation. We designed three specialized annotation agents, each tuned for identifying a specific stakeholder group: (1) HCP Agent using Google Gemini 2.5 Pro (temperature=0.2), chosen for its handling of long, multimodal contexts with medical-lineage evidence; (2) SEng Agent using Grok (temperature=0.2), chosen for its alignment with technical discourse; (3) HCR Agent using GPT (temperature=1.0), chosen for its performance on

well-defined tasks with precise prompts. Each agent received consistent instructions, including: role definition of the target stakeholder group, consistent definitions of the five ethical dimensions, step-by-step analysis instructions (search for self-identification → verify healthcare AI context → identify ethical dimensions → extract themes), and structured JSON output requirements. This identified 4,790 threads (21.5%) that included identifiable stakeholder participation and at least one ethical dimension discussed.

3.4.4 Validation and Qualitative Analysis. We manually validated a random sample of 75 annotated threads to assess annotation quality. We present descriptive statistics on the distribution of ethical dimensions across stakeholder groups. Inferential tests comparing distributions would require independence assumptions that our data violates—threads frequently discuss multiple dimensions simultaneously, and the same user may contribute to multiple threads. Therefore, percentages represent the proportion of threads addressing each dimension independently. We selected high-quality threads as exemplars for qualitative analysis to demonstrate how stakeholders discuss each ethical dimension. We acknowledge that online discussions raise representativeness concerns—Reddit users may not reflect the broader practitioner population, and platform norms shape what is discussed. However, this limitation is partially offset by the scale and ecological validity of naturally occurring discussions. For the complete methodology, see our replication package¹.

3.5 Triangulation Approach

Following the established triangulation framework [37], we designed our own triangulation approach and categorised findings into three types: *Convergent findings*: patterns appearing consistently across three sources, showing high confidence. *Complementary findings*: insights where one source enriches or explains patterns from another. *Unique findings*: concerns appearing from one source, particularly deployment proximal issues from online discussions that remain unaddressed.

¹<https://anonymous.4open.science/r/JAWS-E364/>

The confidence levels are assessed based on cross-source agreement: findings supported by all three sources are classified as high confidence, those supported by two sources as medium confidence, and those from a single source as unique insights requiring further validation.

Based on the triangulated findings, we aim to design an ethics assistant that operationalises these insights into practical developer support. The tool design draws on all three data sources: official guidelines from the systematic mapping study inform the Create and Validate functions, while practitioner experiences from interviews and Reddit discussions inform the Compare function. The current tool design is conceptual (see Section 4). We plan to implement a prototype and evaluate it through user studies with healthcare AI developers to assess whether the tool helps them identify relevant ethical concerns, generate actionable requirements, and validate compliance with regulatory frameworks.

Section 4 presents the preliminary results from applying this triangulation method and the proposed tool design.

4 Results

This section presents preliminary findings from our triangulation approach. As our synthesis is ongoing, we report initial results organized by research questions.

4.1 RQ1 Results

4.1.1 Systematic mapping study findings. Our analysis of 39 primary studies identified ethical concerns distributed across five dimensions. Table 1 summarizes each dimension with its definition and key concerns. The mapping study also identified five mitigation strategy themes: ethical frameworks and interdisciplinary collaboration, accountability and continuous oversight, bias mitigation and fairness, user empowerment and transparency, and enhancing trust and interpretability. However, only 13 of 39 studies conducted empirical evaluation of proposed strategies, the remaining studies proposed conceptual frameworks without validation.

4.1.2 Interview findings. The interviews with 17 practitioners revealed distinct AI use contexts across stakeholder groups. Table 2 summarises the primary AI applications and challenges reported by each group. Participants reported challenges related to model reliability, data quality, human oversight gaps and institutional constraints. As a software engineer noted regarding model reliability: “You cannot take any output from these tools verbatim, you always need to sanity check, validate and verify.”. Another healthcare researcher reported data quality issues: “The large model tried to ‘fill in the gaps’ and ended up generating fabricated data...increased the complexity of our manual review process.” Based on these challenges and ethical priorities, we derived stakeholder-specific If-Then rules that translate practitioner concerns into actionable requirements (e.g., “If using AI for diagnosis, Then require clinician verification before patient communication”).

4.1.3 Online discussion mining findings. From the 22,296 Reddit threads collected, our multi-agent annotation identified 4,790 threads containing ethics-relevant discussions. Table 3 shows the distribution by stakeholder group and ethical dimension.

Table 3: Ethical Dimension Discussion Rates

Dimension	Healthcare Professionals (n=3,387)	Software Engineers (n=665)	Healthcare Researchers (n=738)
Safety	2,536 (74.9%)	388 (58.3%)	487 (66.0%)
Accountability	2,290 (67.6%)	313 (47.1%)	454 (61.5%)
Privacy	981 (29.0%)	300 (45.1%)	266 (36.0%)
Transparency	927 (27.4%)	257 (38.6%)	363 (49.2%)
Bias	920 (27.2%)	132 (19.8%)	230 (31.2%)

Note: Threads are discussed in multiple dimensions.

Healthcare Professionals demonstrated strong focus on safety (74.9%) and accountability (67.6%). Software Engineers demonstrated a more balanced discussion across ethical dimensions. Healthcare Researchers showed higher transparency discussion engagement (49.2%) than other groups.

4.2 RQ2 Results

To compare the ethical concerns identified in RQ1 across the three sources, we analyzed how stakeholders prioritize these concerns, identified patterns across data sources, and derived practical insights for requirements specification.

4.2.1 Stakeholder Priorities. Interview participants ranked the five ethical dimensions by importance (1 = highest priority, 5 = lowest priority). Table 4 presents the mean rankings alongside Reddit discussion rates for comparison.

Table 4: Ethical Dimension Priorities: Interview Rankings vs. Reddit Discussion Rates

Dimension	Interview Ranking (Mean Score)	Reddit Discussion Rate (All Groups)
Privacy	#1 ($\bar{x} = 1.2$)	32.3%
Safety	#2 ($\bar{x} = 2.2$)	71.2%
Bias	#3 ($\bar{x} = 3.3$)	26.8%
Accountability	#4 ($\bar{x} = 3.7$)	63.8%
Transparency	#5 ($\bar{x} = 4.5$)	32.3%

Privacy emerged as the top priority for interview participants. As one healthcare professional explained “In Chinese hospitals, patients’ data is strictly governed. No doctors or nurses are allowed to export sensitive data outside the hospital”

Safety was ranked second in interviews but was the most discussed dimension on Reddit (71.2%). One healthcare professional emphasized data containment “For imported machines and bots, the database should be based locally”

Bias was ranked third but was described as “manageable” by the interviewees.

Accountability was ranked fourth in interviews but was highly discussed on Reddit (63.8%). Healthcare professionals emphasized clinician responsibility “Doctors will be held responsible for any mistakes made by medical bots, as they must review outputs before using them”

Transparency was ranked the lowest. Healthcare professionals stated “We don’t really care about the running logic behind AI bots, doctors don’t have time to dig deeper.”

Table 1: Ethical Dimensions Identified in Systematic Mapping Study

Dimension	Definition	Key Concerns
Safety	Generative AI systems need to operate safely and securely, minimizing risks and issues	Misinformation risks, harmful content generation, system reliability
Privacy	Information and privacy of individuals need to be protected when using generative AI systems	Data protection, consent mechanisms, re-identification risks
Transparency	Operations and decisions of generative AI systems are open and understandable to stakeholders	Model explainability, decision traceability, uncertainty communication
Bias	Biases of the system need to be addressed as they may affect fairness and equity	Algorithmic discrimination, dataset imbalance, demographic disparities
Accountability	AI systems need to be responsible and their actions can be traced and justified	Legal responsibility, audit mechanisms, governance frameworks

Table 2: AI Use Contexts and Challenges by Stakeholder Group (Interviews)

Stakeholder	AI Use Contexts	Key Challenges
Software Engineers (n=6)	Clinical model development; AI tools for domain knowledge; System integration and interoperability	Hallucinations and garbage outputs; Institutional data silos; Compliance requirements
Healthcare Professionals (n=5)	Medical bots for surgery and pharmacy; AI-assisted imaging for CT diagnosis; Medical scribe platforms	Diagnostic inaccuracies; Steep learning curves; Insufficient AI training
Healthcare Researchers (n=6)	Data processing and interpretation; Image segmentation and classification; Smart dataset labeling	Fabricated data; Domain mismatches; Batch processing failures

A notable divergence appears from the results: while privacy was ranked the highest in interviews, safety and accountability were major Reddit discussions. This suggests practitioners cognitively prioritize privacy but discuss safety incidents and accountability questions when issues arise in practice.

4.2.2 Triangulated Findings. Table 5 presents our triangulation matrix, showing how findings align or diverge across the three sources. We classify findings by confidence level based on cross-source agreement.

Table 5: Triangulation Matrix: Cross-Source Findings

Finding	SMS	Interviews	Reddit	Confidence
Privacy and Safety are universally prioritized by all stakeholders	✓ Most covered dimensions	✓ Ranked 1st and 2nd by all groups	✓ Safety most discussed (66–75%)	High
Different stakeholder roles have distinct ethical priorities	✓ Domain-dependent themes	✓ Different challenges by role	✓ Discussion patterns vary by group	High
Ethical frameworks lack practical implementation guidance	✓ 67% strategies lack evaluation	✓ Struggle to operationalize	✓ Deployment frustrations	High
Practitioners deprioritize transparency despite academic emphasis	✓ Frequently proposed	✓ Ranked lowest; time constraints	✓ Lower discussion rates	High
Unclear liability when AI makes clinical errors	○ Discussed abstractly	○ One participant raised concerns	✓ Detailed lawsuit scenarios	Medium
AI holds to higher accuracy standards than human clinicians	× Not addressed	× Not addressed	✓ 90% AI vs 70% human example	Unique
AI may learn biases from specific equipment manufacturers or hospital sites rather than clinical patterns	× Not addressed	× Not addressed	✓ Scanner brands correlating with diagnoses	Unique

Legend: ✓ = Present, ○ = Partially present, × = Absent

Convergent Findings (High Confidence) From the matrix, four findings appeared consistently across all sources:

Privacy and safety are universally prioritized: The systematic mapping study identified *privacy* and *safety* as the most covered dimensions. Interview participants across all stakeholder groups ranked privacy first ($(\bar{x} = 1.2)$) and safety second ($(\bar{x} = 2.2)$). Reddit analysis further show safety as the most discussed dimension, with 66–75% of threads addressing it across all groups. **Different stakeholder roles have distinct ethical priorities:** The mapping study noted domain-dependent ethical themes. Interviews revealed that software engineers focus on technical compliance, healthcare professionals emphasize patient outcomes, and researchers prioritize data integrity. Reddit discussions showed different engagement patterns: healthcare professionals focused on safety and accountability, software engineers engaged more with privacy, and researchers discussed transparency more than other groups.

Ethical frameworks lack practical implementation guidance: The mapping study found that 67% of proposed mitigation strategies lacked empirical evaluation. Interview participants described difficulties translating high-level principles into daily practice. Reddit discussions show frustrations with deploying ethical AI in real clinical settings.

Practitioners deprioritize transparency despite academic emphasis: While the mapping study found transparency frequently proposed as a solution, interview participants ranked it lowest among ethical dimensions. Practitioners viewed understanding AI decision logic as secondary to obtaining reliable outputs. Reddit data further show that the transparency discussion rates (27–49%) substantially lower than safety (58–75%) and accountability (47–68%).

Complementary Findings (Medium Confidence). Unclear liability when AI makes clinical errors: From our results, accountability was only addressed abstractly in the mapping study, and only one interview participant emphasised this concern. However, more concrete scenarios were provided in Reddit discussions. One

healthcare professional (self-identified) stated that *"Supposing there are 2 or 3 big AI models that are used across the country, that company has to bear the malpractice debt for whole country... I can see malpractice lawyers building class action lawsuits for all the people whose cancer was missed by the AI."*

Unique findings. From our results, there are two findings that emerged exclusively from Reddit discussions:

AI held to higher accuracy standards than human clinicians: One software engineer (self-identified) stated that *"If an ML tool is 90% accurate, physicians will say it's clearly not ready. Yet they'll leave out that maybe physicians are only right 70% of the time... Physicians will tell you they've never made a mistake, and patients will tell you otherwise."* This result reveals a double standard in how AI systems are evaluated compared to human clinicians. While AI tools are expected to achieve near-perfect accuracy before deployment, human diagnostic performance is rarely measured or disclosed.

AI may learn equipment/site patterns instead of medical conditions: One healthcare researcher (self-identified) stated that *"Let's say that an MRI scanner manufactured by GE is used in a memory clinic while one manufactured by Siemens is used in an ICU. So an algorithm would associate differences in manufacturers with neurodegenerative qualities and the presence of injury."* This illustrates how AI models can learn false correlations from operational factors rather than genuine clinical factors. If certain equipment is consistently used in specific clinical settings, the model may associate equipment characteristics with diagnoses, due to how data is organized. These biases are harder to detect and require auditing beyond conventional checklists.

4.3 RQ3 Results

Based on our triangulated findings, we aim to design an interactive ethics assistant to support ethical requirements elicitation for healthcare AI. A question arises: why has such a tool not emerged despite years of ethical AI research? Our analysis suggests three contributing factors: (1) 67% of proposed mitigation strategies lack empirical evaluation, with researchers focusing on conceptual frameworks rather than practical tools; (2) ethical concerns are stakeholder-specific, so generic checklists cannot address distinct priorities across roles; and (3) deployment concerns such as liability allocation and equipment-related bias are absent from formal guidelines. Our triangulation approach addresses these gaps by synthesizing academic, interview, and online sources into a unified knowledge base. Figure 2 presents the preliminary interface design with three core functions.

Create Requirements. Developers describe their system context and ethical concerns. The assistant generates specific requirements mapped to official guidelines. For example, a privacy concern about "patient data exposure" generates the requirement "remove patient identifiers from images; store scans in password-protected database," with references to EU AI Act Art. 10 and NIST MAP 1.5.

Validate Requirements. Developers input requirements and select target frameworks. The assistant returns validation status (Valid or Missing) with specific gap identification. For instance, "only radiologists can access patient scans" would be flagged as missing access logging (EU AI Act Article 12) and data retention policies (NIST GOVERN 1.5).

Compare with Real-World Scenarios. Developers input a requirement to receive enhancement suggestions from practitioner experiences. For example, "store CT scans in encrypted database" would surface insights such as "patient data must stay within hospital network" (interviews) and "image files contain hidden identifiers" (online discussions), suggesting an enhanced requirement that combines encryption with on-premises hosting and identifier removal.

5 Discussion

This section reflects on our triangulation approach and discusses implications for RE.

5.1 Triangulation Approach

Our study represents a triangulation approach combining a systematic literature review, practitioner interviews, and online discussion mining to elicit ethical requirements. We discuss several observations from this methodological integration.

Triangulation Reveals Blind Spots in Individual Methods A key observation from our study is that each method has systematic blind spots. Academic literature tends toward idealized frameworks detached from deployment realities. Interviews are shaped by what participants consider appropriate to share in formal settings. Online discussions reveal concerns practitioners voice among peers but may not raise to researchers. Without triangulation, researchers relying on any single method risk incomplete understanding of stakeholder concerns. Our triangulation approach addresses this by cross-validating findings across sources, making blind spots visible and enabling more robust requirements elicitation. This aligns with Runeson and Höst's argument that multi-method approaches strengthen validity in software engineering research [12].

Online Discussions as a Complementary Elicitation Channel Prior work has demonstrated the value of mining app reviews [28], forums [38], and social media [39] for requirements discovery. Our study extends this to ethical requirements in healthcare AI. However, crowd-sourced data introduces noise and representativeness concerns [40]. Our triangulation approach addresses this limitation: online discussions reveal deployment concerns that formal methods miss, while literature review and interviews provide structured validation to filter noise and confirm relevance.

Confidence Levels Help Manage Uncertainty Our triangulation matrix assigned confidence levels based on cross-source agreement: high (all three sources), medium (two sources), and unique (single source). This approach, adapted from evidence synthesis in healthcare research [41], enables differentiated treatment of findings—high-confidence concerns inform baseline requirements, while unique findings warrant further investigation. Although preliminary, this framework demonstrates how triangulation can systematically manage uncertainty in requirements elicitation.

Addressing the Evaluation Gap in AI Ethics Our systematic mapping study revealed that only 33% of studies included practical evaluation; the remainder were purely conceptual. This aligns with critiques that the field is saturated with "checklist ethics" lacking execution mechanisms [7, 42, 43]. Our triangulation approach offers one path forward: by grounding ethical requirements in both official guidelines and practitioner experiences, we bridge the gap between abstract principles and implementable specifications.

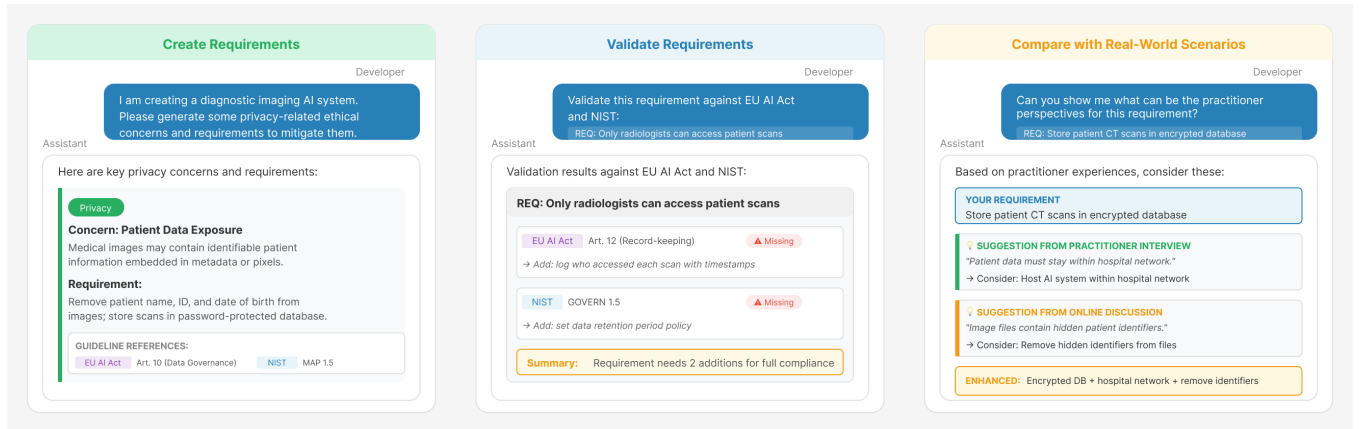


Figure 2: HealthAI Ethics Assistant - Preliminary interface design

5.2 Implications for Software Engineering

Our findings suggest several implications for SE in healthcare AI. **Different stakeholders need different requirements:** Healthcare professionals prioritise diagnostic reliability and accountability, researchers emphasise data quality and validation, and software engineers focus on privacy and system integration. One-size-fits-all ethical checklists are insufficient; developers should tailor elicitation to each role. Our tool’s *Create Requirements* function supports this by generating stakeholder-specific requirements.

Requirements need regulatory traceability: Healthcare AI developers need systematic ways to map requirements to regulatory articles. Our tool’s *Validate Requirements* function addresses this by checking requirements against frameworks like EU AI Act and NIST, identifying gaps before deployment.

Practitioner experience informs better requirements Formal guidelines may miss deployment realities such as hidden identifiers in medical images or liability concerns when AI fails. Our tool’s *Compare with Real-World Scenarios* function surfaces these practitioner insights to enhance requirements.

6 Threats to Validity

Internal Validity: Our interview sample (n=17) was limited to practitioners in China, Australia, and India, potentially missing perspectives from other regulatory contexts, such as the US or the EU. The semi-structured format may have primed participants toward the five predefined ethical dimensions. To mitigate these concerns, we triangulated interview findings with Reddit discussions that include practitioners from diverse geographic contexts. For Reddit analysis, multi-agent LLM annotation may introduce classification errors; although we manually validated a sample, LLM-based annotation remains an active research area with known reliability concerns [44, 45]. We aim to address this in future work by conducting an annotation agreement study with human coders.

External validity: Our findings may not generalise to all healthcare AI contexts. Interviews were limited to practitioners in China, Australia, and India, potentially missing perspectives from other regulatory contexts such as the United States or the European

Union. The Reddit dataset is limited to English-language discussions, underrepresenting practitioners from non-English-speaking healthcare systems where regulatory and cultural contexts may differ. We address these limitations through triangulation, which identifies findings consistent across diverse data sources. We will expand our sample to additional regions and explore multilingual online communities as we develop our ethics assistant.

7 Conclusion

This paper investigated how to bridge AI ethics and requirements engineering in healthcare by applying *methodological triangulation* to the elicitation of ethical requirements. We integrated evidence from a systematic mapping study, semi-structured practitioner interviews, and large-scale mining of online discussions, treating these sources as a unified study rather than independent snapshots. The results show why this integration matters: triangulation surfaces concerns that are either muted or absent when relying on a single method, and it enables explicit reasoning about which candidate requirements are broadly supported versus context-specific.

Across sources, we observed (i) *convergent* concerns such as privacy and safety that consistently appear as stakeholder priorities, (ii) *complementary* insights that show stakeholder-specific requirements and trade-offs, and (iii) *unique* deployment issues, including equipment-related bias and asymmetric evaluation standards that are difficult to get from literature or interviews alone.

As next steps, we will (a) replicate the triangulation pipeline in additional high-stakes domains beyond healthcare to assess transferability and uncover domain-specific ethical requirements, (b) evaluate the ethics assistant with software engineers in realistic requirements workflows to measure its utility for elicitation, prioritisation, and trade-off justification, and (c) extend the knowledge base longitudinally by incorporating new evidence sources (e.g., post-deployment incident reports and regulatory updates) to study how ethical requirements evolve over time. Overall, triangulation provides a principled way to build and maintain ethical requirements knowledge as AI systems and regulatory expectations change, while keeping the supporting evidence transparent for auditing and review.

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